

TYRONE MARHGUY

Computer Engineering | Digital Hardware & Computer Systems

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SUMMARY

Computer Engineering junior at the University of Pennsylvania focused on digital hardware systems spanning computer architecture, RTL/FPGA design, ASIC implementation, and verification. Built a custom 32-bit computer and timing-closed low-latency FPGA systems; merged fixes into OpenROAD, Verilator, and LibreLane.

EDUCATION

University of Pennsylvania

B.S.E. Computer Engineering

Philadelphia, PA

Aug. 2024 – May 2028

Coursework: Computer Organization & Design, VLSI Fundamentals, Digital Logic, Circuit-Level Modeling, Embedded Systems

SELECTED HARDWARE PROJECTS

Tomato – Custom 32-bit Computer Architecture | GitHub

2026 – Present

- Architected a custom 32-bit ISA/CPU around a three-source Dual-LUT ALU; implemented the complete machine on Nexys A7, booting an assembly-written OS with HDMI output, input, and games; built the assembler and browser ISA emulator.
- Validated the datapath across Verilator, UVM/SVA, and SymbiYosys formal; synthesized it in Sky130 and fabricated/soldered an 8-bit 74xx Dual-LUT ALU slice.

64-bit RISC-V CPU Core | GitHub

2025 – 2026

- Designed and verified a synthesizable **RV64IM 5-stage pipeline** in SystemVerilog with forwarding, load-use stalls, variable-latency Wishbone memory, M-mode traps/CSRs, and branch prediction; Verilator/cocotb regression passes the RV32IM compliance suite.

Low-Latency FPGA Networking & NASDAQ ITCH | UDP/IP | ITCH

2026

- Built a pure-RTL RMII/MAC/IPv4/UDP stack on Artix-7 with no soft CPU/OS in the hot path; closed timing at **100 MHz** (WNS +**1.985 ns**) with **20 ns** simulated cut-through payload-echo latency.
- Built live-Ethernet ITCH 5.0 parsing and a dual-sided 512-slot BRAM order book; closed timing at **100 MHz**, with **30 ns** parse and **60 ns** Add-to-BBO latency; ran a **10M-message** Verilator/cocotb regression with zero BBO mismatches.

ASIC Datapaths & Memory | BF16 MAC | 6T SRAM

2026

- Designed a 2-stage BF16 → FP32 MAC with a 4-cycle streaming interface; verified 1,000 randomized vectors against a Python/cocotb golden model and built a local LibreLane/Sky130 implementation flow.
- Designed a full-custom **16 x 4 6T SRAM** in PTM 22 nm HP with StrongARM sensing, replica-bitline timing, and automated NGSpice timing/power sweeps; demonstrated **4.571 GHz** sustained simulated operation.

ENGINEERING EXPERIENCE & OPEN SOURCE

Vero Electric

Hardware & Firmware Engineer

Philadelphia, PA

May 2026 – Aug. 2026

- Designed/routed a BQ79616 16-channel battery-monitoring front end for a **240 kWh / 120 kW** energy-storage system; closed PCB ERC/DRC with zero errors.
- Developed embedded C/C++ fault-detection and SPI/I²C acquisition firmware; supported board bring-up with oscilloscopes and logic analyzers.

Aragorn AI

Software Engineer Intern

Remote

May 2026 – Aug. 2026

- Contributed to production backend/API-integration workflows for AI services; debugged cross-service issues and documented service interfaces for production integrations.

Open-Source EDA Contributor – OpenROAD, Verilator, LibreLane

2026

- Merged upstream C++/Python fixes for OpenROAD LEF58 parsing, Verilator Linux peak-memory reporting, and LibreLane/Yosys synthesis compatibility/check reporting, with regression coverage; LibreLane fixes shipped in releases.

TECHNICAL SKILLS

Architecture / RTL: SystemVerilog, Verilog; CPU/SoC microarchitecture, datapaths, pipelining, memory interfaces, CDC

Verification: UVM, SVA, cocotb, Verilator, SymbiYosys formal, Questa, golden-model/regression testing

FPGA / ASIC / VLSI: Vivado, Artix-7, Yosys, OpenROAD, LibreLane/OpenLane, Sky130, synthesis, STA, P&R, SPICE

Programming / Hardware: Python, C/C++, Tcl, Bash, Linux, Git; Ethernet/RMII, UDP, SPI, I²C, UART